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4.2.1 The four fundamental spaces of a matrix

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by a vector row
vector, then what
you're doing is you're
taking linear

combinations of the rows of A . That of course is a row vector. The row space

of a consists of
column vectors that
can be created by
taking linear
combinations

of the rows of A
when you view those
rows as column
vectors. That's what
this

here says and
obviously the row
space of A is equal to
the column space of
the



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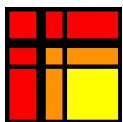
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Reading assignment

1/1 point (graded)

Read Unit 4.2.1 of the notes. [[LINK](#)]



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Homework 4.2.1.1

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$. Show that its row space, $\mathcal{R}(A)$, and null space, $\text{Null}(A)$, are orthogonal.

☒ done



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✓ Correct (1/1 point)

Homework 4.2.1.2

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$. Show that its column space, $\mathcal{C}(A)$, and left null space, $\text{Null}(A^H)$, are orthogonal.

☒ done



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✓ Correct (1/1 point)

Homework 4.2.1.3

1/1 point (graded)
Let $\{s_0, \dots, s_{r-1}\}$ be a basis for subspace $\mathcal{S} \subset \mathbb{C}^n$ and $\{t_0, \dots, t_{k-1}\}$ be a basis for subspace $\mathcal{T} \subset \mathbb{C}^n$. Show that the following are equivalent statements:

- Subspaces $\mathcal{S}, \mathcal{T}$ are orthogonal.
- The vectors in $\{s_0, \dots, s_{r-1}\}$ are orthogonal to the vectors in $\{t_0, \dots, t_{k-1}\}$.
- $s_i^H t_j = 0$ for all $0 \leq i < r$ and $0 \leq j < k$.
- $\left(\begin{array}{c|c|c} s_0 & \cdots & s_{r-1} \end{array} \right)^H \left(\begin{array}{c|c|c} t_0 & \cdots & t_{k-1} \end{array} \right) = 0$, the zero matrix of appropriate size.

☒ done



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Homework 4.2.1.4

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$. Show that any vector $x \in \mathbb{C}^n$ can be written as

Let $A \in \mathbb{C}^{m \times n}$. Show that any vector $x \in \mathbb{C}^n$ can be written as $x = x_r + x_n$, where $x_r \in \mathcal{R}(A)$ and $x_n \in \text{Null}(A)$, and $x_r^H x_n = 0$.

☒ done



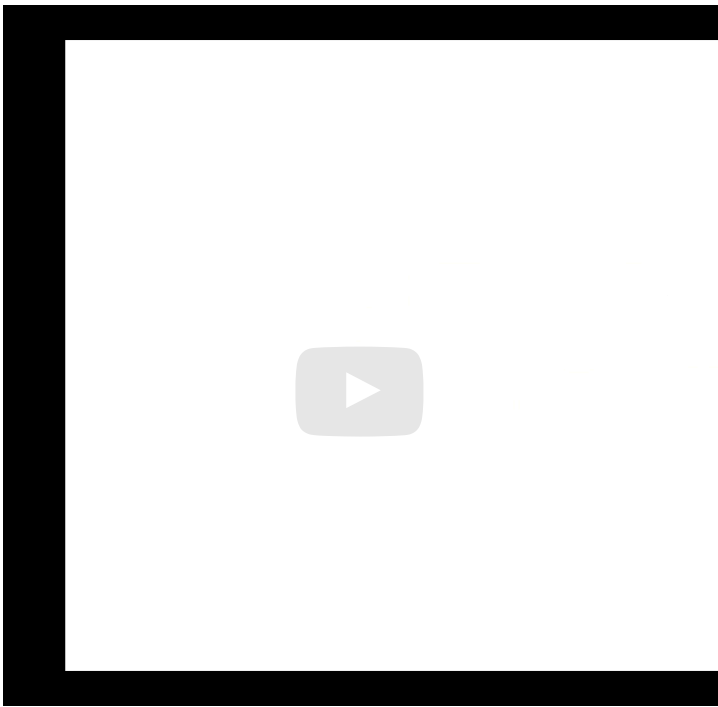
Solution:

For a complete solution, see [the notes](#).

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 Answers are displayed within the problem

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that the vector x can be uniquely written as the sum of a vector in the row space, x_r , and a vector in the null space, x_n . And we know that the vector x_n maps to the zero vector in $\mathbb{C}^m$. Alright, so if we look at A times x then we see that that's A times x_r plus x_n . And that means that it's A times x_r , plus A times x_n . And we know that A times x_n is 0 . So we get that that's equal to A times x_r . So what that really means is that

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